PROPERTIES OF CONTEXT-FREE LANGUAGES

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Non-terminals play variables

LEMMA

Let $\mathcal{G}$ be a grammar and let $\mathcal{G}'$ be obtained from $\mathcal{G}$ by refreshing all of its non-terminals with new symbols. Then $\mathcal{L}(\mathcal{G}') = \mathcal{L}(\mathcal{G})$.

Non-terminals play variables

EXAMPLE

- For instance, take the grammar

$$\begin{array}{rcl} S & \rightarrow & aSbC \mid ab \\ C & \rightarrow & cC \mid c \end{array}$$

  and the renaming $[R/S, D/C]$.
- Obtain the grammar

$$\begin{array}{rcl} R & \rightarrow & aRbD \mid ab \\ D & \rightarrow & cD \mid c \end{array}$$

- Every derivation step for the first grammar can be mimicked for the second grammar, and viceversa

Closure wrt union

LEMMA

The class of free languages is closed w.r.t. set-union.

- Meaning:
  - If $\mathcal{L}_1$ and $\mathcal{L}_2$ are free languages
  - Then $\mathcal{L}_1 \cup \mathcal{L}_2$ is a free language

Closure wrt union

PROOF

- Let $\mathcal{L}_1$ and $\mathcal{L}_2$ be free languages.
- Then there exist two free grammars $\mathcal{G}_1 = (V_1, T_1, S_1, \mathcal{P}_1)$ and $\mathcal{G}_2 = (V_2, T_2, S_2, \mathcal{P}_2)$ such that $\mathcal{L}_1 = \mathcal{L}(\mathcal{G}_1)$ and $\mathcal{L}_2 = \mathcal{L}(\mathcal{G}_2)$
- Also, we can assume that $\mathcal{G}_1$ and $\mathcal{G}_2$ do not share any non-terminal namely that $(V_1 \setminus T_1) \cap (V_2 \setminus T_2) = \emptyset$
- Let $\mathcal{G} = (V_1 \cup V_2 \cup \{S\}, T_1 \cup T_2, S, \mathcal{P}_1 \cup \mathcal{P}_2 \cup \{S \to S_1 \mid S_2\})$ where:
  - S is a new symbol not in $V_1 \cup V_2$
- Then $\mathcal{L}(\mathcal{G})$ is free and $\mathcal{L}(\mathcal{G}) = \mathcal{L}(\mathcal{G}_1) \cup \mathcal{L}(\mathcal{G}_2)$

Closure wrt union

PROOF: Why is $\mathcal{G}$ free?

- The productions of $\mathcal{G}$ are those in $\mathcal{P}_1 \cup \mathcal{P}_2 \cup \{S \to S_1 \mid S_2\})$
- The productions in both $\mathcal{P}_1$ and $\mathcal{P}_2$ have the form $A \to \alpha$
- The productions $S \to S_1$ and $S \to S_2$ also have the form $A \to \alpha$
- Hence $\mathcal{G}$ is free

Closure wrt union

PROOF: Why is $\mathcal{L}(\mathcal{G}) = \mathcal{L}(\mathcal{G}_1) \cup \mathcal{L}(\mathcal{G}_2)$?

- $w \in \mathcal{L}(\mathcal{G})$
- iff there exists a derivation $S \Rightarrow^* w$
- iff $S \Rightarrow S_1 \Rightarrow^* w$ or $S \Rightarrow S_2 \Rightarrow^* w$
- iff $S_1 \Rightarrow^* w$ or $S_2 \Rightarrow^* w$
- iff $w \in \mathcal{L}(\mathcal{G}_1)$ or $w \in \mathcal{L}(\mathcal{G}_2)$
- iff $w \in \mathcal{L}(\mathcal{G}_1) \cup \mathcal{L}(\mathcal{G}_2)$

Closure wrt union

PROOF: Why insisting on $\mathcal{G}_1$ and $\mathcal{G}_2$ not sharing non-terminals?

- Take
$$\mathcal{G}_1 : \begin{array}{lcl} S_1 & \rightarrow & aA \\ A & \rightarrow & a \end{array} \qquad\qquad \mathcal{G}_2 : \begin{array}{lcl} S_2 & \rightarrow & bA \\ A & \rightarrow & b \end{array}$$
- Then $\mathcal{L}(\mathcal{G}_1) = \{aa\}$ and $\mathcal{L}(\mathcal{G}_2) = \{bb\}$.
- However, if we just put everything together and define
$$\mathcal{G} : \begin{array}{lcl} S & \rightarrow & S_1 \mid S_2 \\ S_1 & \rightarrow & aA \\ S_2 & \rightarrow & bA \\ A & \rightarrow & a \mid b \end{array}$$
- Then $\mathcal{L}(\mathcal{G}) = \{aa, ab, ba, bb\} \neq \mathcal{L}(\mathcal{G}_1) \cup \mathcal{L}(\mathcal{G}_2)$.
- What is the problem here?

Closure wrt union

PROOF: Why insisting on $\mathcal{G}_1$ and $\mathcal{G}_2$ not sharing non-terminals?

- Take

$$\mathcal{G}_1: \begin{array}{lcl} S_1 & \to & aA \\ A & \to & a \end{array} \qquad \mathcal{G}_2: \begin{array}{lcl} S_2 & \to & bA \\ A & \to & b \end{array}$$

- Then $\mathcal{L}(\mathcal{G}_1) = \{aa\}$ and $\mathcal{L}(\mathcal{G}_2) = \{bb\}$.
- The symbol A plays distinct roles in $\mathcal{G}_1$ and in $\mathcal{G}_2$
- To reflect this distinction, rather take

$$\mathcal{G}_1: \begin{array}{lcl} S_1 & \to & aA \\ A & \to & a \end{array} \qquad \mathcal{G}_2: \begin{array}{lcl} S_2 & \to & bA' \\ A' & \to & b \end{array}$$

- No mix up of productions now, and, by construction $\mathcal{L}(\mathcal{G}) = \{aa, bb\}$

Closure wrt concatenation

LEMMA

The class of free languages is closed w.r.t. concatenation

- Meaning:
  - If $\mathcal{L}_1$ and $\mathcal{L}_2$ are free languages
  - Then $\{w_1 w_2 \mid w_1 \in \mathcal{L}_1 \text{ and } w_2 \in \mathcal{L}_2\}$ is a free language

Closure wrt concatenation

PROOF

- Let $\mathcal{L}_1$ and $\mathcal{L}_2$ be free languages
- Then there exist two free grammars $\mathcal{G}_1 = (V_1, T_1, S_1, \mathcal{P}_1)$ and $\mathcal{G}_2 = (V_2, T_2, S_2, \mathcal{P}_2)$ such that $\mathcal{L}_1 = \mathcal{L}(\mathcal{G}_1)$ and $\mathcal{L}_2 = \mathcal{L}(\mathcal{G}_2)$
- Also, we can assume that there is no clash between the non-terminals of $\mathcal{G}_1$ and those of $\mathcal{G}_2$
- Let $\mathcal{G} = (V_1 \cup V_2 \cup \{S\}, T_1 \cup T_2, S, \mathcal{P}_1 \cup \mathcal{P}_2 \cup \{S \to S_1 S_2\})$ where S is a new symbol not in $V_1 \cup V_2$
- Then $\mathcal{L}(\mathcal{G})$ is free and $\mathcal{L}(\mathcal{G}) = \{w_1 w_2 \mid w_1 \in \mathcal{L}(\mathcal{G}_1) \text{ and } w_2 \in \mathcal{L}(\mathcal{G}_2)\}$

Are free languages closed w.r.t. intersection?

To say something about this issue we will soon introduce the Pumping Lemma for free languages

Before that, we see how grammars can be manipulated to meet specific constraints on productions

Chomsky normal form

LEMMA

Let $\mathcal{L}$ be a context-free language. Then there exists a context-free grammar $\mathcal{G}$ such that $\mathcal{L}(\mathcal{G}) = \mathcal{L} \setminus \{\epsilon\}$ and $\mathcal{G}$ is such that

1. It has no ϵ-production (i.e. no production of the shape $A \to \epsilon$)
2. It has no unit production (i.e. no production of the shape $A \to B$)
3. It has no useless non-terminal (i.e. non-terminals that never appear in some derivations of some strings of terminals)
4. Each of its productions have
   - either the form $A \to a$
   - or the form $A \to A_1 A_2$

Chomsky normal form

ELIMINATIONS AT WORK

$$\begin{array}{lcl} S & \to & ABC \mid abc \\ A & \to & aB \mid \epsilon \\ B & \to & bA \mid C \\ C & \to & \epsilon \end{array}$$

Chomsky normal form

ϵ-PRODUCTION ELIMINATION

- Find **nullable** non-terminals, i.e., every non-terminal A such that $A \Rightarrow^* \epsilon$
- It is an inductive process
  - **Base:** if $A \to \epsilon$ is a production, then A is nullable
  - **Iteration:** if $A \to Y_1 Y_2 \ldots Y_n$ is a production and $Y_1, Y_2, \ldots, Y_n$ are all nullable, then A is nullable
- Substitute each production $A \to Y_1 Y_2 \ldots Y_n$ by a family of productions where combinations of nullable Y_is are removed from the body (Exception: If all of Y_is are nullable do not take in the family the production $A \to \epsilon$)
- Eliminate every production $A \to \epsilon$

Chomsky normal form

ϵ-PRODUCTION ELIMINATION

$$
\begin{array}{lcl}
S & \to & ABC \mid abc \\
A & \to & aB \mid \epsilon \\
B & \to & bA \mid C \\
C & \to & \epsilon
\end{array}
$$

- Find **nullable** non-terminals
  - A and C nullable by $A \to \epsilon, C \to \epsilon$
  - B nullable by $B \to C$ and C nullable
  - S nullable by $S \to ABC$ and A, B, C nullable

Chomsky normal form
ϵ-PRODUCTION ELIMINATION

$$
\begin{array}{lcl}
S & \rightarrow & ABC \mid abc \\
A & \rightarrow & aB \mid \epsilon \\
B & \rightarrow & bA \mid C \\
C & \rightarrow & \epsilon
\end{array}
$$

- Substitute each production $A \rightarrow Y_1 Y_2 \ldots Y_n$ by a family of productions (if all of Y_is are nullable is useless to take in the family the production $A \rightarrow \epsilon$)

$$
\begin{array}{lcl}
S & \rightarrow & ABC \mid abc \mid AB \mid AC \mid BC \mid A \mid B \mid C \\
A & \rightarrow & aB \mid a \mid \epsilon \\
B & \rightarrow & bA \mid b \mid C \\
C & \rightarrow & \epsilon
\end{array}
$$

Chomsky normal form
ϵ-PRODUCTION ELIMINATION

$$
\begin{array}{lcl}
S & \rightarrow & ABC \mid abc \mid AB \mid AC \mid BC \mid A \mid B \mid C \\
A & \rightarrow & aB \mid a \mid \epsilon \\
B & \rightarrow & bA \mid b \mid C \\
C & \rightarrow & \epsilon
\end{array}
$$

- Eliminate every production $A \rightarrow \epsilon$

$$
\begin{array}{lcl}
S & \rightarrow & ABC \mid abc \mid AB \mid AC \mid BC \mid A \mid B \mid C \\
A & \rightarrow & aB \mid a \\
B & \rightarrow & bA \mid b \mid C
\end{array}
$$

Chomsky normal form

USELESS NON-TERMINAL ELIMINATION

$$\begin{array}{lcl} S & \rightarrow & ABC \mid abc \mid AB \mid AC \mid BC \mid A \mid B \mid C \\ A & \rightarrow & aB \mid a \\ B & \rightarrow & bA \mid b \mid C \end{array}$$

- Useless non-terminal elimination (Here C is useless: No strings of terminals can be derived from any string containing C)

$$\begin{array}{lcl} S & \rightarrow & abc \mid AB \mid A \mid B \\ A & \rightarrow & aB \mid a \\ B & \rightarrow & bA \mid b \end{array}$$

Chomsky normal form

UNIT PRODUCTION ELIMINATION

$$\begin{array}{lcl} S & \rightarrow & abc \mid AB \mid A \mid B \\ A & \rightarrow & aB \mid a \\ B & \rightarrow & bA \mid b \end{array}$$

- Unit production elimination

$$\begin{array}{lcl} S & \rightarrow & abc \mid AB \mid aB \mid a \mid bA \mid b \\ A & \rightarrow & aB \mid a \\ B & \rightarrow & bA \mid b \end{array}$$

Chomsky normal form

UNIT PRODUCTION ELIMINATION

- From

$$\begin{array}{rcl} S & \rightarrow & ABC \mid abc \\ A & \rightarrow & aB \mid \epsilon \\ B & \rightarrow & bA \mid C \\ C & \rightarrow & \epsilon \end{array}$$

- We get

$$\begin{array}{rcl} S & \rightarrow & abc \mid AB \mid aB \mid a \mid bA \mid b \\ A & \rightarrow & aB \mid a \\ B & \rightarrow & bA \mid b \end{array}$$

Chomsky normal form

ONCE ALL ELIMINATIONS ARE DONE

After eliminating

- ϵ-productions
- unit productions
- useless non-terminals

We get a grammar whose productions have the form $A \rightarrow \beta$ where

- either β is a single terminal
- or β is such that $|\beta| \geq 2$

Now transform the grammar to get productions with

- either the form $A \rightarrow a$
- or the form $A \rightarrow A_1 A_2$

Chomsky normal form

LATEST TRANSFORMATION: AN EXAMPLE

Example

- $S \rightarrow aSb \mid ab$
- Both aSb and ab are to be changed. Pick up a new non-terminal for a, say A, and a new non-terminal for b, say B, and transform into
- $S \rightarrow ASB \mid AB, \qquad A \rightarrow a, \qquad B \rightarrow b$
- ASB is longer than 2, pick up a new non-terminal for SB, say C, and transform into

  $S \rightarrow AC \mid AB, \qquad C \rightarrow SB, \qquad A \rightarrow a, \qquad B \rightarrow b$

Pumping Lemma for free languages

LEMMA

Let $\mathcal{L}$ be a free language. Then

- $\exists p \in \mathbb{N}^+$ such that
- $\forall z \in \mathcal{L}$ such that $|z| > p$
- $\exists u, v, w, x, y$ such that
  - $z = uvwxy$ and
  - $|vwx| \leq p$ and
  - $|vx| > 0$ and
  - $\forall i \in \mathbb{N}.uv^iwx^iy \in \mathcal{L}$

Pumping Lemma for free languages
PROOF

- Let $\mathcal{L}$ be a free language
- The lemma is about words longer than $p > 0$, and hence different from ϵ
- Then just consider a free grammar $\mathcal{G}$ in Chomsky normal form such that $\mathcal{L}(\mathcal{G}) = \mathcal{L} \setminus \{\epsilon\}$

Pumping Lemma for free languages
PROOF

- Let k be the number of non-terminals in $\mathcal{G}$
- Observe that any derivation tree for words in $\mathcal{L}(\mathcal{G})$ has
  - Exactly 2^0 nodes at level 0
  - At most 2^1 nodes at level 1
  - At most 2^2 nodes at level 2
  - ......
  - At most 2^j nodes at level j
- Take $p = 2^{k+1}$

Pumping Lemma for free languages

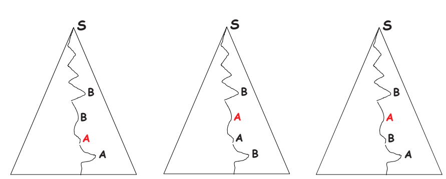
PROOF

- Let $z \in \mathcal{L}$ be such that $|z| > p$
- Then the derivation tree for z must have at least $k + 2$ levels
- Then every longest path of the derivation tree for z has a terminal at the last level and traverses at least $k + 1$ non-terminals
- Hence there is at least one non-terminal occurring twice or more along the path

Pumping Lemma for free languages

PROOF

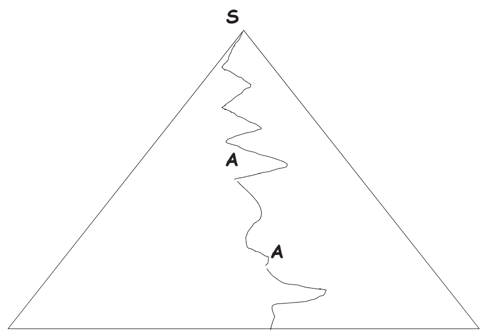
- Consider the longest path in the tree, and the **deepest pair** of occurrences of the same non-terminal along that path (i.e. choose the non-terminal whose second occurrence is found first going bottom-up)
- For instance, below the deepest pair is always the pair of As



Pumping Lemma for free languages

PROOF

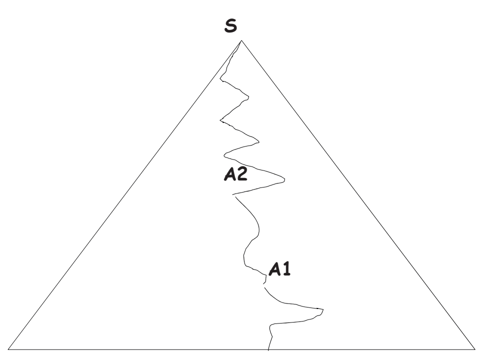
Let A be the non-terminal of the deepest pair of occurrences of the same non-terminal along the path



Pumping Lemma for free languages

PROOF

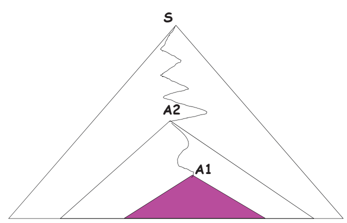
Call $A1$ and $A2$ the two occurrences of A

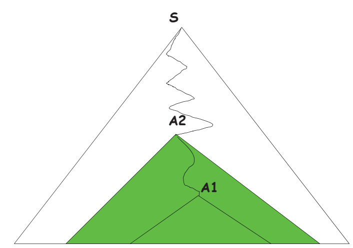


Pumping Lemma for free languages

PROOF

Then there are two distinct subtrees rooted at A: the "pink subtree" and the "green subtree"





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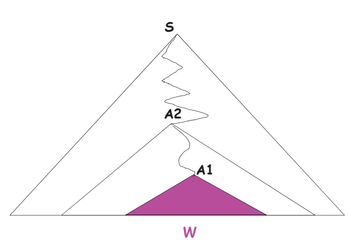
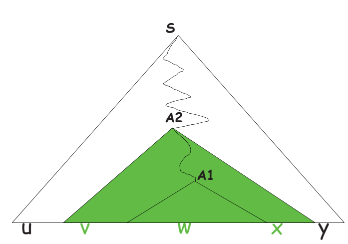
Pumping Lemma for free languages

PROOF

Then $z = uvwxy$

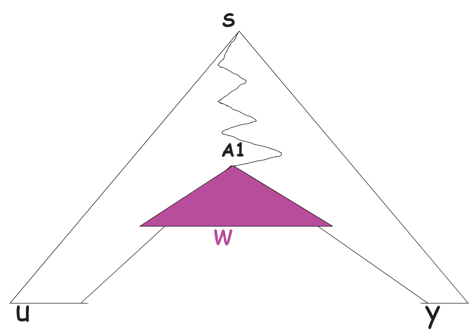




Pumping Lemma for free languages

PROOF

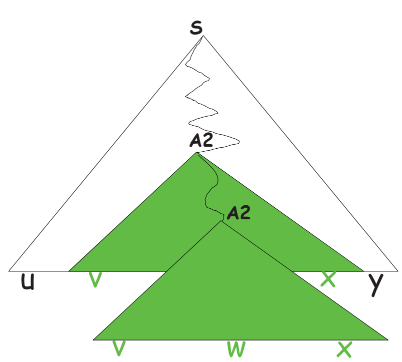
Then $uv^0wx^0y \in \mathcal{L}$.



Pumping Lemma for free languages

PROOF

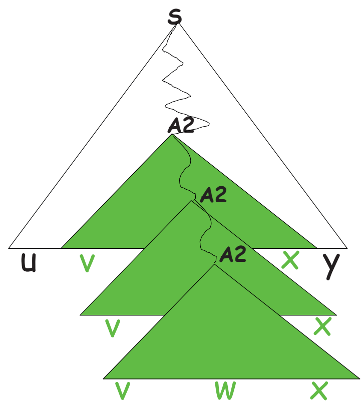
Then $uv^2wx^2y \in \mathcal{L}$.



Pumping Lemma for free languages

PROOF

Then $uv^3wx^3y \in \mathcal{L}$.



Pumping Lemma for free languages

PROOF

Then

- $\forall i \in \mathbb{N}. uv^iwx^iy \in \mathcal{L}$
- $|vwx| \leq p$
  - Because we have chosen the deepest pair of repeated non-terminals $(A1, A2)$ along the longest path from S
  - Then along the longest path from $A2$, below $A2$, no non-terminal can occur more than once
  - Then the subtree rooted at $A2$ has at most $k + 1$ levels
  - Then the length of the yield of the subtree is bound by 2^{k+1}
- $|vx| > 0$
  - Because $\mathcal{G}$ is in Chomsky normal form
  - Then it cannot be $A \Rightarrow^+ A$. It can only be $A \Rightarrow^+ \alpha A \beta$ with at least one terminal derived by either α or β

Pumping Lemma for free languages

WHAT IS THIS LEMMA GOOD FOR?

- Recall the structure of the statement
- " Let $\mathcal{L}$ be a free language. Then *PL-THESIS*."
- **By no means** the lemma can be used to show that a certain language is free
- It is rather used to show that a **language is not free**
- Schema of such proofs:
  - Assume that language $\mathcal{L}$ is free
  - Show that *PL-THESIS* is false, i.e., prove *not(PL-THESIS)*
  - By contradiction, conclude that $\mathcal{L}$ is not free

Pumping Lemma for free languages

PL-THESIS

- $\exists p \in \mathbb{N}^+ .\ \forall z \in \mathcal{L}\colon\ |z| > p.\ \exists u, v, w, x, y.\ P$
- where
  - $P \equiv P1$ and $P2$ and $P3$ and $P4$
  - $P1 \equiv z = uvwxy$
  - $P2 \equiv |vwx| \leq p$
  - $P3 \equiv |vx| > 0$
  - $P4 \equiv \forall i \in \mathbb{N}. uv^i wx^i y \in \mathcal{L}$

Pumping Lemma for free languages

not(PL-THESIS)

- not ($\exists p \in \mathbb{N}^+.\ \forall z \in \mathcal{L}\colon\ |z| > p.\ \exists u, v, w, x, y.\ P$)
- $\forall p \in \mathbb{N}^+.$ not ($\forall z \in \mathcal{L}\colon\ |z| > p.\ \exists u, v, w, x, y.\ P$)
- $\forall p \in \mathbb{N}^+.\ \exists z \in \mathcal{L}\colon\ |z| > p.$ not ($\exists u, v, w, x, y.\ P$)
- $\forall p \in \mathbb{N}^+.\ \exists z \in \mathcal{L}\colon\ |z| > p.\ \forall u, v, w, x, y.$ not (P)
- $\forall p \in \mathbb{N}^+.\ \exists z \in \mathcal{L}\colon\ |z| > p.\ \forall u, v, w, x, y.$
  not ($P1$ and $P2$ and $P3$ and $P4$)

Pumping Lemma for free languages

not ($P1$ and $P2$ and $P3$ and $P4$)

- not ($P1$ and $P2$ and $P3$ and $P4$)
- not (($P1$ and $P2$ and $P3$) and $P4$)
- not ($P1$ and $P2$ and $P3$) or not $P4$
- ($P1$ and $P2$ and $P3$) implies not $P4$
- ($P1$ and $P2$ and $P3$) implies not ($\forall i \in \mathbb{N}. uv^iwx^iy \in \mathcal{L}$)
- ($P1$ and $P2$ and $P3$) implies $\exists i \in \mathbb{N}.$ not ($uv^iwx^iy \in \mathcal{L}$)

Pumping Lemma for free languages

not(PL-THESIS)

- $\forall p \in \mathbb{N}^+.\ \exists z \in \mathcal{L}\colon\ |z| > p.\ \forall u, v, w, x, y.$
  - ($z = uvwxy$ and $|vwx| \leq p$ and $|vx| > 0$)
  - implies
  - $\exists i \in \mathbb{N}. uv^i wx^i y \notin \mathcal{L}$

- Operationally:

- **Whichever** positive natural number p is
- Carefully **choose** a word z longer than p and belonging to the language such that
- **Whichever** unpacking of z into $uvwxy$ with $|vwx| \leq p$ and $|vx| > 0$ is taken
- A natural number i can be **chosen** which is such that $uv^i wx^i y \notin \mathcal{L}$

Pumping Lemma at work

$$\begin{array}{rrcl} \mathcal{G}: & S & \rightarrow & aSBc \mid abc \\ & cB & \rightarrow & Bc \\ & bB & \rightarrow & bb \end{array}$$

- $\mathcal{G}$ is context-dependent and $\mathcal{L}(\mathcal{G}) = \{a^n b^n c^n \mid n > 0\}$.
- Is $\mathcal{L}(\mathcal{G})$ a free language?

Pumping Lemma at work

LEMMA

$\mathcal{L} = \{a^n b^n c^n \mid n > 0\}$ is not free

Pumping Lemma at work

PROOF

- Suppose $\mathcal{L}$ is free, and let p be an arbitrary positive integer
- Take $z = a^p b^p c^p$
- If ($z = uvwxy$ and $|vwx| \leq p$ and $|vx| > 0$)
- Then vx cannot have occurrences of both as and cs because the last occurrence of a and the first occurrence of c are $p + 1$ positions far. In fact, for some positive k and j
  - Either $vwx = a^k$
  - Or $vwx = a^k b^j$
  - Or $vwx = b^j$
  - Or $vwx = b^j c^k$
  - Or $vwx = c^k$

Pumping Lemma at work

PROOF

- Then vx has either no occurrences of c or no occurrences of a
- Then uv^0wx^0y cannot have the form $a^nb^nc^n$, hence $uv^0wx^0y \notin \mathcal{L}$
- Then, by contradiction wrt the Pumping Lemma, $\mathcal{L}$ is not free

Pumping Lemma at work

AGAIN ON THE PROOF STRUCTURE

- ... let p be an arbitrary positive integer
  $\forall p$: Any p
- Take $z = a^pb^pc^p$
  $\exists z$: Choose $z \in \mathcal{L}$ longer than p
- ... If ($z = uvwxy$ and $|vwx| \leq p$ and $|vx| > 0$) then ....
  $\forall u, v, w, x, y : z = uvwxy$ and $|vwx| \leq p$ and $|vx| > 0$
- ... $uv^0wx^0y \notin \mathcal{L}$
  $\exists i$: Choose a value for the iterator
- ... contradiction

Pumping Lemma at work

$$
\begin{array}{llcl}
\mathcal{G}: & S & \rightarrow & CD \\
 & C & \rightarrow & aCA \mid bCB \mid \epsilon \\
 & AD & \rightarrow & aD \\
 & BD & \rightarrow & bD \\
 & Aa & \rightarrow & aA \\
 & Ab & \rightarrow & bA \\
 & Ba & \rightarrow & aB \\
 & Bb & \rightarrow & bB \\
 & D & \rightarrow & \epsilon
\end{array}
$$

- $\mathcal{G}$ is context-dependent
- What is $\mathcal{L}(\mathcal{G})$?

Pumping Lemma at work

- Derived strings have a bookmark D initially at rightmost position

$$S \rightarrow CD$$

- Strings can only grow longer by replacing C

$$C \rightarrow aCA \mid bCB \mid \epsilon$$

- Non-terminals close to the rightmost delimiter can be converted to the corresponding terminal

$$
\begin{array}{lcl}
AD & \rightarrow & aD \\
BD & \rightarrow & bD
\end{array}
$$

- Non-terminals and terminals can be swapped when the terminal is at the right of the non-terminal

$$
\begin{array}{lcl}
Aa & \rightarrow & aA \\
Ab & \rightarrow & bA \\
Ba & \rightarrow & aB \\
Bb & \rightarrow & bB
\end{array}
$$

Pumping Lemma at work

$$
\begin{array}{ll}
S & \text{by } S \to CD \\
\Rightarrow CD & \text{by } C \to aCA \\
\Rightarrow aCAD & \text{by } C \to aCA \\
\Rightarrow aaCAAD & \text{by } C \to bCB \\
\Rightarrow aabCBAAD & \text{by } C \to \epsilon \\
\Rightarrow aabBAAD & \text{by } AD \to aD \\
\Rightarrow aabBAaD & \text{by } Aa \to aA \\
\Rightarrow aabBaAD & \text{by } Ba \to aB \\
\Rightarrow aabaBAD & \text{by } AD \to aD \\
\Rightarrow aabaBaD & \text{by } Ba \to aB \\
\Rightarrow aabaaBD & \text{by } BD \to bD \\
\Rightarrow aabaabD & \text{by } D \to \epsilon \\
\Rightarrow aabaab &
\end{array}
$$

Pumping Lemma at work

$$
\begin{array}{rrcl}
\mathcal{G}: & S & \to & CD \\
& C & \to & aCA \mid bCB \mid \epsilon \\
& AD & \to & aD \\
& BD & \to & bD \\
& Aa & \to & aA \\
& Ab & \to & bA \\
& Ba & \to & aB \\
& Bb & \to & bB \\
& D & \to & \epsilon
\end{array}
$$

- $\mathcal{L}(\mathcal{G}) = \{ww \mid w \in \{a, b\}^*\}$
- Is $\mathcal{L}(\mathcal{G})$ a free language?

Pumping Lemma at work

LEMMA
$\mathcal{L}(\mathcal{G}) = \{ww \mid w \in \{a,b\}^*\}$ is not free

Proof
Analogous to the previous one.
A good choice for z is $z = a^p b^p a^p b^p$

Free or not?

- $\{a^n b^n c^n \mid n > 0\}$
- Not free, by previous lemma

- $\{a^n b^n c^j \mid n, j > 0\}$
- Free, concatenation of two free languages
- $\{a^n b^n \mid n > 0\}$ and $\{c^j \mid j > 0\}$

- $\{a^j b^n c^n \mid j, n > 0\}$
- Free, concatenation of two free languages
- $\{a^j \mid j > 0\}$ and $\{b^n c^n \mid n > 0\}$

Closure wrt intersection does not hold

LEMMA
The class of free languages is not closed w.r.t. intersection.

Proof

- By contradiction
- Take two free languages $\mathcal{L}_1$ and $\mathcal{L}_2$ whose intersection is not free
- $\mathcal{L}_1 = \{a^n b^n c^j \mid n, j > 0\}$
- $\mathcal{L}_2 = \{a^j b^n c^n \mid n, j > 0\}$
- $\mathcal{L}_1 \cap \mathcal{L}_2 = \{a^n b^n c^n \mid n > 0\}$

Training

$$\begin{aligned} \mathcal{G}: \quad S &\rightarrow aSc \mid aTc \mid T \\ T &\rightarrow bTa \mid ba \end{aligned}$$

- Is $\mathcal{G}$ ambiguous?
- Yes
  - $S \Rightarrow aTc \Rightarrow abac$
  - $S \Rightarrow aSc \Rightarrow aTc \Rightarrow abac$
- What is $\mathcal{L}(\mathcal{G})$?

  $\{a^k b^n a^n c^k \mid k \geq 0, n > 0\}$

Training

$$\mathcal{G}: \begin{array}{lcl} S & \to & 0B \mid 1A \\ A & \to & 0 \mid 0S \mid 1AA \\ B & \to & 1 \mid 1S \mid 0BB \end{array}$$

- What is $\mathcal{L}(\mathcal{G})$?

  $\{w \mid w \in \{0,1\}^* \text{ and } \#(0,w) = \#(1,w)\}$

Training

- Define $\mathcal{G}$ such that $\mathcal{L}(\mathcal{G}) = \{a^k b^n c^{2k} \mid k, n > 0\}$

$$\begin{array}{lcl} S & \to & aScc \mid aBcc \\ B & \to & bB \mid b \end{array}$$

- Define $\mathcal{G}$ such that $\mathcal{L}(\mathcal{G}) = \{a^k b^n c^{2k} \mid k, n \geq 0\}$

$$\begin{array}{lcl} S & \to & aScc \mid B \\ B & \to & bB \mid \epsilon \end{array}$$

Training

$$\mathcal{G}: \quad \begin{array}{rcl} S & \rightarrow & aBS \mid bA \\ aB & \rightarrow & Ac \mid a \\ bA & \rightarrow & S \mid Ba \end{array}$$

- Is $\mathcal{L}(\mathcal{G}) = \emptyset$?

  No: $\underline{S} \Rightarrow aB\underline{S} \Rightarrow \underline{aB}bA \Rightarrow a\underline{bA} \Rightarrow \underline{aB}a \Rightarrow aa$

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Training

- Define a grammar $\mathcal{G}$ such that $\mathcal{L}(\mathcal{G})$ is the set of all the even binary numbers

  $$S \rightarrow 0S \mid 1S \mid 0$$

- Define a grammar $\mathcal{G}'$ such that $\mathcal{L}(\mathcal{G}') = \{1^n0 \mid n \geq 0\}$

  $$S \rightarrow 1S \mid 0$$

- Is $\mathcal{L}(\mathcal{G}') = \mathcal{L}(\mathcal{G})$?

  No: 000000 is in $\mathcal{L}(\mathcal{G})$ but not in $\mathcal{L}(\mathcal{G}')$